Parameter-Efficient Adaptation of SAM 2 for Sentinel-1 SAR Flood Inundation Mapping in the Indo-Gangetic Region

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Why this paper exists

- Bangladesh floods every monsoon. *Cloud cover* makes optical satellites unreliable when flood maps matter most.
- Sentinel-1 SAR **penetrates clouds**, available globally and free.
- But SAR is a noisy, single-channel, logarithmic-intensity signal, nothing like the natural RGB photographs vision foundation models (SAM, SAM 2) are pretrained on.
- **Question:** can we adapt SAM 2 to radar, *cheaply*, and have it generalize to an unseen flood event?

Answer (preview): yes, mostly, with one important caveat about the cross-polarization channel.

Three gaps in prior work

1. **No head-to-head comparison** of parameter-efficient fine-tuning (PEFT) methods on SAM-family backbones for SAR floods. Prior papers use one method each; you can't tell which adapter to pick.

2. **No test of operational utility** of uncertainty estimates. Prior SAR-flood work reports aggregate calibration (low ECE) but does not test whether the resulting *per-pixel* scores are useful for triage.

3. **No public reproducible OOD test set** combining temporal *and* geographic out-of-distribution shift on Sentinel-1. Standard benchmarks hold out one country, but all events are from 2017–2019.

We close all three.

Methodology: a controlled empirical sweep

5 backbones $\times$ 4 PEFT methods $\times$ 3 seeds

Backbone	LoRA	DoRA	Conv-LoRA	AdaptFormer
SAM ViT-B (94 M)	✓	✓	✓	✓
SAM 2 Hiera-B+ (81 M)	✓	✓	✓	✓
SAM ViT-L (308 M)			✓	
SAM ViT-H (636 M)			✓	
SAM 2 Hiera-L (224 M)			✓	

- Full sweep + baselines: $\approx$ **3.5 hours**, $\approx$ **\$10** cloud compute (dual NVIDIA RTX PRO 5000 Blackwell).
- Uncertainty: **Monte Carlo dropout**, 20 stochastic passes per chip.

Polarimetric “pseudo-RGB” input

SAM 2 was pretrained on three RGB channels. Sentinel-1 gives us only two:

- **VV** (co-polarization): vertical transmit, vertical receive.
- **VH** (cross-polarization): vertical transmit, horizontal receive.

We must synthesize a third channel.

Three compositions compared:

ratio	$(\,\mathrm{VV},\ \mathrm{VH},\ 10\log_{10}(P_{VV}/P_{VH})\,)$
diff	$(\,\mathrm{VV},\ \mathrm{VH},\ \mathrm{VV}_{dB} - \mathrm{VH}_{dB}\,)$
single	$(\,\mathrm{VV},\ \mathrm{VV},\ \mathrm{VV}\,)$

By the dB-domain identity $10\log_{10}(P_{VV}/P_{VH}) = \mathrm{VV}_{dB} - \mathrm{VH}_{dB}$, *ratio* and *diff* produce identical model inputs. The meaningful contrast is between dual-polarization (ratio or diff) and single-polarization (VV alone).

Four evaluation splits

Split	Chips	Role
Sen1Floods11 test	90	In-distribution (10 countries)
Sen1Floods11 Bolivia	15	Held-out country
Pakistan-S1F11 subset	12	Regional in-distribution (2010 event)
Pakistan-2022	**39**	**Strict OOD: new geography + time**

- The first three splits come from **Sen1Floods11** (Bonafilia et al., CVPR Workshops 2020). All events are from 2017–2019.
- The fourth, **Pakistan-2022**, is constructed in this work. It covers the August–September 2022 Indo-Gangetic monsoon flood, the same meteorological system that produces Bangladesh's annual flooding.

Pakistan-2022 dataset construction

Two public sources, paired in this work:

- **Labels:** TU Wien Sentinel-1-derived flood-extent product (Roth et al., NHESS 2023; CC-BY 4.0)
- **Imagery:** Microsoft Planetary Computer `sentinel-1-rtc` (Radiometric Terrain Corrected, anonymous SAS-token access)

Pipeline: for each TU Wien mask date, find the matching Sentinel-1 RTC scene, read at native 10 m, reproject the mask upward to 10 m, tile into 512×512 chips.

Result: 39 chips at native 10 m, fully Sen1Floods11-compatible layout.

Acquisition note: the first chip set was at 20 m and mismatched the training distribution. We rebuilt at native 10 m; that is the dataset released here.

Experimental setup

Training protocol:

- 10 epochs on the 252 Sen1Floods11 hand-labelled training chips.
- AdamW optimizer; separate learning rates 10^{-4} (adapters) / 10^{-5} (unfrozen pretrained weights); cosine schedule with 200-step linear warmup.
- Loss: Dice + binary cross-entropy with logits, equal weight.
- Mixed-precision fp16; batch size 1 with gradient accumulation 4 (effective batch 4).
- **No training-time data augmentation.** Test-time augmentation (4 rotations $\times$ 2 flips $=$ 8 views) is applied only at inference for the ensemble condition.

Seeds: 42, 123, 20025 (three independent reruns per cell).

The next half: what we actually found.

Across the four evaluation splits, plus one diagnostic finding we did *not* expect:

The cross-polarization channel turns out to be the operative out-of-distribution failure mode.

Headline results

Method	Sen1F11 test	Bolivia	Pak-S1F11	Pak-2022
U-Net baseline	0.601	**0.684**	0.271	**0.672**
Hiera-B+ + AdaptFormer	**0.634**	0.458	0.254	0.090
Hiera-B+ + LoRA	0.599	0.635	0.255	0.140
Hiera-B+ + DoRA	0.602	0.637	0.254	0.132

- **In-distribution:** AdaptFormer wins (0.634), beating U-Net at a fraction of trainable parameters.
- **Bolivia held-out:** DoRA matches U-Net (0.637 vs 0.684).
- **Pakistan-2022:** every SAM-family cell collapses to 0.09–0.14, while U-Net retains 0.672.

The Pakistan-2022 collapse: not a backbone or adapter problem

- Every SAM-family cell collapses, regardless of:
  - Backbone (ViT-Base, ViT-Large, ViT-Huge, Hiera-Base-Plus, Hiera-Large, all collapse)
  - PEFT method (LoRA, DoRA, Conv-LoRA, AdaptFormer, all collapse)
  - Random seed
- U-Net on the same training data does not collapse. So the collapse is *specific to the adapted SAM-family models*, not a property of the dataset.
- **Question:** what is the SAM family seeing that the U-Net is not?

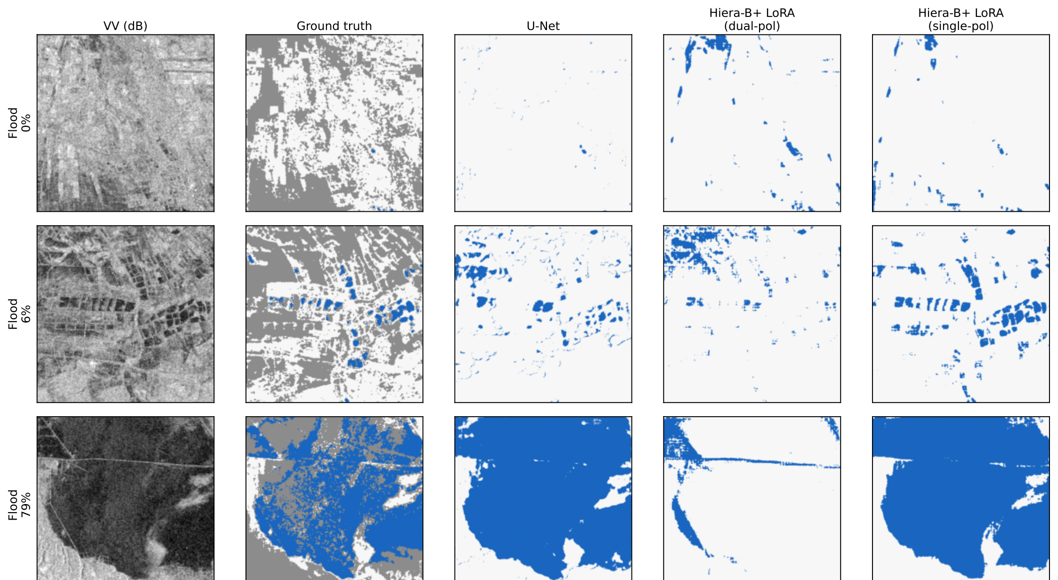
The polarimetric ablation: cross-polarization is the culprit

Pakistan-2022 IoU	Dual-polarization	Single-polarization (VV only)
Hiera-B+ Conv-LoRA	0.113	$\mathbf{0.632 \pm 0.057}$
Hiera-B+ LoRA	0.144	$\mathbf{0.631 \pm 0.050}$
Hiera-B+ DoRA	0.131	$\mathbf{0.658 \pm 0.014}$
Hiera-B+ AdaptFormer	0.092	$\mathbf{0.556 \pm 0.050}$
U-Net reference	0.672	

Drop the cross-polarization (VH) channel and the collapse vanishes.

- Single-polarization Pakistan-2022 IoU jumps to 0.56–0.66 across three seeds.
- Within 0.01–0.12 of the U-Net baseline.
- **Three-seed evidence; no overlap with the dual-pol band.**

Qualitative comparison: same chip, three predictions



Why? The transformer amplifies cross-polarization distribution shift

- The cross-polarization channel *helps* in-distribution. Its marginal statistics on Pakistan-2022 differ from Sen1Floods11.
- Possible drivers: different scattering geometry, different surface conditions, different acquisition mode parameters.
- The SAM transformer's global attention *amplifies* this distribution shift.
- The U-Net's local receptive fields *tolerate* it.

Operational recommendation: for new regions or events, abstain from the cross-polarization channel. Use VV only.

Negative finding: MC dropout is not per-pixel useful on OOD

Aggregate calibration is fine:

- ECE $= 0.053$ on in-distribution test (at the well-calibrated threshold).
- ECE $= 0.067$ on Pakistan-2022 (essentially unchanged despite the IoU collapse).

But pointwise selective-prediction curves are flat:

- Abstaining on the most uncertain pixels does *not* improve IoU on the remaining ones.
- Aggregate calibration $\not\Rightarrow$ per-pixel difficulty ranking on OOD events.
- Contradicts the implicit assumption in prior SAR-flood uncertainty work.

Three gaps, three closures

1. **PEFT comparison:** controlled head-to-head; accuracy-versus-generalization trade-off documented. AdaptFormer maximizes in-distribution IoU; LoRA family is more generalization-robust.
2. **Calibration vs operational utility:** substantive negative finding. Aggregate ECE does not imply per-pixel uncertainty utility on OOD.
3. **Public OOD test set:** Pakistan-2022 (39 chips, native 10 m) released with acquisition pipeline.

Bonus diagnostic finding: cross-polarization is the operative OOD failure mode for SAM-family adapters.

Limitations and future work

Honest limitations:

- Pakistan-2022 evaluation rests on 39 chips and TU Wien labels derived from a Sentinel-1 model.
- Stretch backbones evaluated only under Conv-LoRA.
- Bangladesh-Sylhet construction pipeline is released, chips are not yet built or evaluated.

Future work:

- Empirical evaluation on the Bangladesh-Sylhet test set.
- Characterize the cross-polarization shift directly (incidence angle, acquisition mode).
- Sharper uncertainty estimators: deep ensembles, learned reject heads.

Thank you